

Interval Matching on the Circle

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We assuming $(L_i, R_i) \in [0, 2\pi)$ and always consider the counterclockwise arc.

Let $\{L_i\}$ be the set of all left endpoints. Note that $L_i \in [0, 2\pi)$. For each L_i , we unenroll the circle onto the line:

$$R_j \xrightarrow{\alpha} \begin{cases} R_i & \text{if } R_i < L_j \text{ and } L_i < R_j, \\ R_j & \text{otherwise.} \end{cases}$$

Run the regular algorithm on each of these instances; if any of these instances finds a matching then output the matching, otherwise indicate none exist.

We prove correctness now. It suffices to show that whenever such a matching exists, our algorithm also finds a matching. Let $\{P_i\}$ denote this matching with $P_i \in [0, 2\pi]^2 \times [0, 2\pi]$. Fix a cyclic ordering $P_1 < \dots < P_n$.

Perform our transformation using $P_1 = ((L_1, R_1), x_1)$. It suffices to argue that $\{P_i\}$ corresponds to a matching $\{\alpha(P_i)\}$. By correctness of our regular algorithm, we know that we must identify a matching too then.